FrugalReason Master Evaluation Report (qwen2.5:3b)

1. Executive Summary & Core Results

This report details the final 16-cell strict evaluation run comparing FrugalReason v3 against 3 standard baselines across 4 benchmark datasets using the qwen2.5:3b model.

2. Performance Metrics by Dataset

Dataset: AQUA

Strategy	Accuracy	Avg Tokens	Avg Latency	Avg Calls	Parse Rate
greedy_cot	66.0%	570	0.0s	1.0	100.0%
self_consistency_k5	70.0%	2853	0.0s	5.0	100.0%
best_of_n_k5_self_eval	62.0%	5703	0.0s	10.0	96.0%
frugal_reason_v3	82.0%	3505	44.5s	5.0	100.0%

Dataset: GSM_HARD

Strategy	Accuracy	Avg Tokens	Avg Latency	Avg Calls	Parse Rate
greedy_cot	0.0%	536	0.0s	1.0	100.0%
self_consistency_k5	0.0%	2694	0.0s	5.0	100.0%
best_of_n_k5_self_eval	0.0%	5446	0.0s	10.0	100.0%
frugal_reason_v3	0.0%	3944	50.5s	6.6	100.0%

Dataset: MATH

Strategy	Accuracy	Avg Tokens	Avg Latency	Avg Calls	Parse Rate
greedy_cot	0.0%	509	0.0s	1.0	100.0%
self_consistency_k5	0.0%	2536	0.0s	5.0	100.0%
best_of_n_k5_self_eval	0.0%	5099	0.0s	10.0	100.0%
frugal_reason_v3	0.0%	2834	38.4s	5.2	100.0%

Dataset: SVAMP

Strategy	Accuracy	Avg Tokens	Avg Latency	Avg Calls	Parse Rate
greedy_cot	4.0%	273	0.0s	1.0	76.0%
self_consistency_k5	0.0%	1449	0.0s	5.0	100.0%
best_of_n_k5_self_eval	2.0%	2952	0.0s	10.0	76.0%
frugal_reason_v3	0.0%	2476	37.1s	8.8	100.0%

3. GATE Check (FrugalReason v3 vs. Greedy CoT)

Dataset	Greedy Acc	Frugal Acc	Accuracy Diff	Frugal Parse Rate
AQUA	66.0%	82.0%	+16.0%	100.0%
GSM_HARD	0.0%	0.0%	+0.0%	100.0%
MATH	0.0%	0.0%	+0.0%	100.0%
SVAMP	4.0%	0.0%	-4.0%	100.0%